CENTERED NUCLEAR DUALITY, ADDITION-AWARE PINNING, AND RELATIVE-MASS OBSTRUCTIONS IN MONOTONE BOOLEAN CIRCUIT LOWER BOUNDS

ARGUS OPEN RESEARCH PROJECT

Abstract. In the study of monotone and positive-weight Boolean circuit lower bounds for fixed-factor approximation of $R_n = 2F_G - T_G$ over the full Boolean cube, we establish a two-sided equivalence between constant-factor circuit approximation bounds and a centered nuclear-error dual charge, prove an addition-aware pinning decomposition and fixed-tolerance signed-test transfer reduction, and construct explicit $O(M_n + N^2)$-gate nonnegative rational circuits and affine-line incidence matrices over $\mathbb{F}_{4^\ell}$ refuting uncentered relative-mass, naive rarity-budget, and sampler-free matrix-gluing shortcuts.

1. Main Mathematical Results

Theorem 1.1 (Centered nuclear-error duality, addition-aware pinning, and shortcut refutations)**.** *Approximating $R_n = 2F_G - T_G$ within a fixed factor $\kappa_0 > 1$ on the Boolean cube under the canonical sampler is equivalent up to absolute constants to bounding the centered nuclear-error dual charge and reduces via addition-aware pinning to the outer-sampler signed test, whereas uncentered relative-mass and sampler-free matrix-gluing relaxations are refuted by explicit $O(M_n + N^2)$-gate and $\mathbb{F}_{4^\ell}$ affine-plane constructions.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3395 in the domain of computational complexity and circuit lower bounds adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Two-sided equivalence between fixed-factor Boolean cube approximation and centered nuclear-error charge, addition-aware pinning decomposition, signed-test transfer reduction, and explicit O(M_n+N^2)-gate refutations of uncentered relative-mass and matrix-only gluing shortcuts.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted P vs. NP separation

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 3395 (computational complexity and circuit lower bounds), archived in the accompanying repository package `results/3395/`.

Date: November 4, 2026.

2020 *Mathematics Subject Classification.* Primary 68Q17; Secondary 94C10, 06E30.